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SYSTEMATIC REVIEW

Effectiveness of Education Interventions Designed to Improve Nutrition Knowledge in Athletes: A Systematic Review

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Abstract

Background A range of nutrition education strategies are used to assist athletes to improve nutrition knowledge. Evaluation of nutrition education interventions guides the delivery of efficacious nutrition education for athletes.

Aim Our aim was to systematically review sport/general nutrition education interventions delivered to athletes, and to evaluate their effectiveness.

Methods A search was conducted using terms related to nutrition knowledge, athletes, education, and intervention. Included studies had to be conducted in athletes (all calibres), use a scored nutrition knowledge assessment tool, and measure knowledge before and after a nutrition education intervention. Peer-reviewed and unpublished theses were included.

Results Thirty-two manuscripts (randomised controlled trial, $n = 13$; single-arm pre/post design, $n = 19$) met the inclusion criteria. Participants ($n = 2180$; mean age 17.4 ± 1.7 years) were mostly female (66.1%), university-level (56.3%) athletes based in the US (75%). Comparison of different education methods was limited since the majority of interventions ($n = 28/36$, 77.8%) used face-to-face education. Most intervention conditions ($n = 19$) had a total contact time of < 300 min (5 h). The majority of interventions ($n = 30$, 85.7%) reported significant knowledge improvement, with a mean increase of 16.1 ± 0.7% (mean effect size 1.72; range 0.4–17.1). Only 15.6% of studies used well-validated knowledge assessment tools (more than three types of validity or reliability testing).

Conclusions Most interventions reported a significant improvement in nutrition knowledge. Unfortunately, the wide range of knowledge assessment tools with limited validation, and the predominant use of face-to-face interventions, prohibits identification of the most effective modality and dose for nutrition education in athletes.

1 Introduction

Nutrition education interventions are specific programmes designed to assist targeted populations in acquiring relevant knowledge or adopting improved eating behaviours [1, 2]. In athlete settings, these sport/general nutrition interventions have the goal of maintaining or improving health and enhancing athletic performance. Food selection is affected by a wide range of factors, including taste, cost, convenience and knowledge [3]. An athlete's food choices may be further affected by factors such as time constraints, financial insecurity and the frequency of travel associated with sporting commitments [4, 5].

Athletes participating in elite sports typically train in an environment where the knowledge and skills needed to optimise sport performance are taught, yet studies report these athletes still have suboptimal diets and low levels of nutrition knowledge [6–10]. As with the general population, athletes are exposed to conflicting nutrition information from friends, family, team-mates, and the internet [11]. Athletes with access to sports nutritionists as key sources of information have been reported to adhere more closely to established nutrition principles despite heavy marketing of popular diets and dietary supplements promising

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Key Points

Nutrition education interventions mostly led to an improvement in athletes' nutrition knowledge.

There is still a need for a well-validated, reliable and comprehensive nutrition knowledge assessment instrument.

Future studies should consider the use of technology-based education or currently underutilised methods such as cooking classes or supermarket tours in conjunction with a robust study design.

improvement in performance, physique, recovery or well-being which can challenge evidence-based nutrition recommendations [12, 13]. Coaches and trainers are often relied upon to provide nutrition information where there is limited access to a sports nutritionist; however, they are also reported to have suboptimal sport/general nutrition knowledge [14–17]. Given the tight-knit nature of sporting environments, the perpetual spread of nutrition misinformation can be difficult to prevent; however, knowledge of sport nutrition is essential before an athlete can select the most appropriate foods for both their sport and for health [12, 18, 19].

Structured nutrition education is usually provided to athletes through their sports programmes using a range of modalities, including face-to-face individual consultations, group education sessions, interactive activities (e.g. cooking classes, grocery store tours), and technology-based interventions [20]. However, the effectiveness of different types of nutrition education methods for disseminating sport/general nutrition knowledge in athletes has not yet been evaluated.

2 Objectives

Sport/general nutrition education programmes aimed at athletes need to be evaluated for their effectiveness in changing nutrition knowledge. Programme evaluation allows facilitators to identify the most successful educational methods for changing sport/general nutrition knowledge, while providing insight into nutrition concepts athletes find difficult to comprehend. Therefore, the aim of this review was to evaluate the effectiveness of different sport/general nutrition education interventions targeted at improving knowledge in athletes. A secondary aim was to determine the educational method and intervention duration most effective in improving nutrition knowledge. Evaluation of nutrition education interventions provides insight for future development of initiatives to support the delivery of efficacious nutrition education to athletes.

3 Methods

3.1 Design

The search was conducted by one author (RT) using the MEDLINE (Ovid), SPORTDiscus (EBSCO), CINAHL (EBSCO), Web of Science, SCOPUS and ProQuest Central electronic databases from earliest record until 9 June 2019.

The search terms used were 'nutrition knowledge', 'diet knowledge', 'food knowledge', 'health knowledge', 'dietary behavior*', 'dietary behaviour*' 'health literacy', 'nutrition literacy', 'food', 'nutri*', 'diet*', 'supplement*', 'diet* supplement', 'competition eating', 'hydration', 'weight management', 'label reading', 'nutrition information panel', 'nutrition label*', 'food label*' combined with 'elite sportsperson*', 'competitive athlete*', 'professional athlete*', 'athlete*', 'Olympic', 'sport/s*' and 'diet* education', 'nutrition* education', 'health education', 'evaluat*', 'survey*', 'screen*', 'intervention', 'nutrition* intervention', 'diet* intervention', 'health intervention', 'dietary intake', 'nutrition program*', 'behavior change', 'behaviour change' 'knowledge*', 'measure', 'question*', 'assess*'. The bibliographies of papers included were hand searched for any articles missed during the database search. The review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement [21].

3.2 Inclusion and Exclusion Criteria

Inclusion criteria were (1) original research conducted in humans; (2) interventions aimed at improving sport/general nutrition knowledge in athletes; (3) use of a nutrition knowledge assessment instrument producing a quantitative score pre/post intervention; and (4) availability in English. Athletes were defined as those regularly engaged in sports at any level (recreational through to elite), all ages, male and female, and included those with physical (not intellectual) disabilities. If the inclusion criteria were met, theses and dissertations were included. Research studies were excluded if nutrition knowledge levels were self-rated, targeted military or police forces, or investigated education topics beyond the scope of nutrition, such as the use of banned performance-enhancing substances (e.g. anabolic steroids). Conference posters and abstracts were also excluded.

After removing duplicates, the remaining articles were screened by one author (RT.). Articles that could not be excluded by title or abstract were retrieved in full and

independently reviewed by one author (RT) to determine their eligibility for inclusion.

3.3 Data Extraction

Data extractions were conducted independently on each paper by two of three authors (RT, KB, MM). The names of the authors, year of publication, country where the study was conducted, and study design were recorded. Demographic factors extracted included athlete calibre, sport played, sex, mean age and age range or standard deviation (SD). Information about the nutrition knowledge instrument used were recorded and included any details of instrument validation, question types, scoring, and topics assessed. Intervention information was also extracted and included the curriculum, mode of delivery, duration, attendance/attrition, and pre- and post-intervention nutrition knowledge scores. Where an article contained insufficient detail, attempts were made to obtain missing information by contacting the authors.

3.4 Assessment of Methodological Quality

One author (RT) assessed all studies and the second review was conducted by one of two other authors (KB and JG). A modified Downs and Black checklist for assessment of methodological quality was used (Electronic Supplementary Tables S1 and S2) [22]. In modifying the checklist, 24 of 27 items were retained, while items 8, 13 and 17 were removed as they did not apply to the study designs of the included papers. Items 14, 15, 21, 22, 23 and 24 were not deemed relevant for single-arm study designs. Item 20, "Was the tool used to measure nutrition knowledge accurate (valid and reliable)?", was modified to allow a maximum score of two instead of one. This allowed reviewers to differentiate between papers that conducted two or less validity or reliability measures of their nutrition knowledge instrument compared with others that used multiple (three or more) validity or reliability tests. Two additional items from the Academy of Nutrition and Dietetics (AND) quality criteria checklist [23] were added; specifically, item 9, "Are conclusions supported by results with biases and limitations taken into consideration?"; item 10, "Is bias due to study's funding or sponsorship unlikely?". Finally, items 3.4, 5.1, 5.2, 7.2, 7.3, 7.4, 7.5, 7.6, 7.7, 9.1 and 9.2 from the AND checklist were considered concurrently with similar Downs and Black items to assist reviewers in their quality ratings. The maximum scores were 22 and 28 for the single-arm and two-arm studies, respectively. Internal (inter-rater) validity was checked across items for each paper. Differences in rating scores were discussed among authors to achieve consensus.

3.5 Data Analysis

When sport/nutrition knowledge scores were reported as raw data they were converted to a percentage to facilitate comparison between manuscripts. Average age, attrition (when reported) and changes in knowledge scores were calculated using a weighted mean. Hedges' g (random model) for effect size (ES) was calculated where a mean and SD were reported. The extracted data (mean, sample size and SD) were transferred to Comprehensive Meta-Analysis (CMA) version 2 (Biostat, 2005, Englewood, IL, USA) to calculate the ES. For single-arm studies, paired sample data were used to calculate the ES. ESs were interpreted using Cohen's cut points (small effect: 0.2; medium effect: 0.5; large effect: > 0.8) [24]. Due to the heterogeneity in instruments used to assess sport/nutrition knowledge and intervention durations, meta-analysis was not conducted.

4 Results

The search identified 69,226 potential manuscripts, and an additional 19 potentially relevant manuscripts were identified by hand-searching. After the removal of duplicates, 51,213 manuscripts remained. A total of 50,902 records were excluded by title only. The remaining 311 manuscripts were screened by abstract, of which 68 full-text articles were retrieved. Of these, 32 met all the inclusion criteria. A summary of the PRISMA process combined with the search is shown in Fig. 1. The randomised controlled trial (RCT) by Rennolds [25] tested three arms (presentation only, handout only, and presentation combined with handout) against a no-intervention control group (Table 1). Where reported, intervention groups were almost always compared with no intervention. In the RCT by Karpinski [26], the control group received general nutrition handouts instead of the intervention. Within the single-arm studies (Table 2), the study by Gould tested two experimental conditions (cooking lesson combined with handouts vs. handouts only) [27], and one study contained two population groups and reported results separately (adults and teenagers) [28]. Within the 32 included manuscripts, there were a total of 36 experimental conditions.

4.1 Randomised Controlled Trials

Overall, 41% (13/32 studies) of the manuscripts used an RCT design (Table 1). Of these, seven were published in peer-reviewed journals and the remainder were dissertations or theses ($n = 6$). Age was reported in nine studies (weighted mean 17.3 ± 1.4 years) [26, 29–36]. Across all studies, the total number of athletes recruited was 1529, of

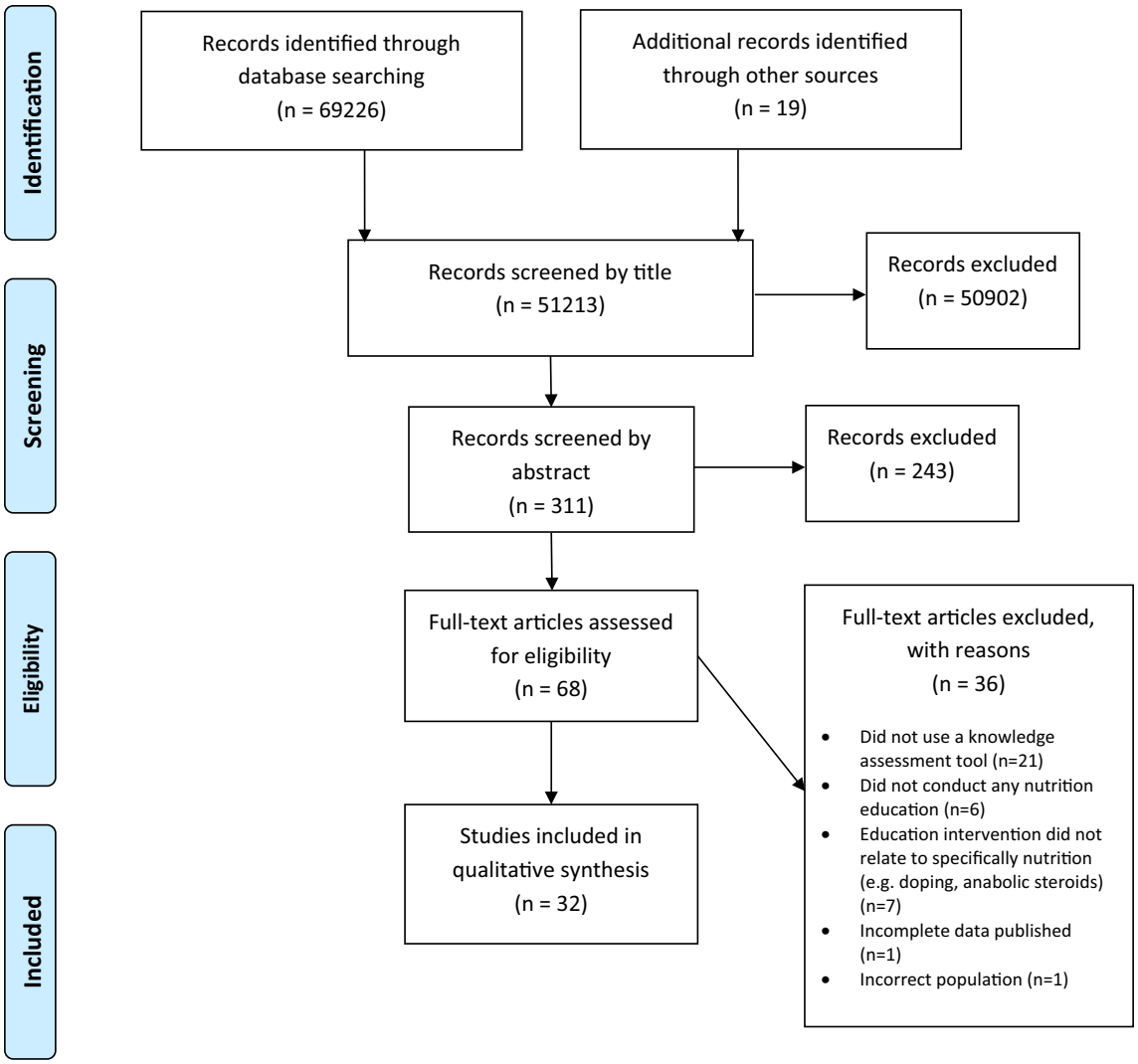

Fig. 1 PRISMA flow chart showing the process for including studies

whom 75.8% ($n = 1159$) were female; the mean attrition rate was 16.2 ± 2.5% (range 0–70.3%) (Table 3).

The majority of the RCTs (12/13 studies; 92.3%) were conducted in the US. Of these studies, the majority ($n = 7$) involved college-level athletes (53.8%) [25, 26, 29–31, 37, 38], with three studies specifically targeting athletes competing at the Division 1 level (the highest level of college athletes, from colleges with the largest athletic budgets and number of scholarships available for student-athletes) [25, 29, 30]. Only one study included elite-level athletes competing professionally at a national or international level [35]. Sports most frequently represented were soccer (football; $n = 5$) [29, 31, 33, 36, 38], followed by swimming ($n = 4$) [29, 30, 34, 38] and track/athletics ($n = 4$) [30, 31, 37, 38]; however, 14.5% ($n = 222$) of participants did not have a sport designation given. Soccer players represented almost half (41.5%) of all athletes ($n = 635$), however the majority ($n = 535$) were from one study [36].

Knowledge assessment instruments were primarily author-designed (8/13 studies; 61.5%) [25, 29–32, 34, 37, 38] or were modified from a pre-existing instrument (4/13 studies; 30.1%) [26, 35, 36, 39]. Adaptations of existing instruments were mainly undertaken to ensure relevance to the study population. Pilot testing was used as a form of validity/reliability testing in 61.5% of studies (8/13 studies) [26, 29, 30, 32, 33, 35, 38, 39]. Other forms of validation/reliability testing included test–retest ($n = 4$) [29, 30, 32, 35], content validation with an expert panel ($n = 3$) [29–31], Cronbach α for internal consistency ($n = 3$) [32, 35, 36] and assessing readability ($n = 2$) [29, 30]. Two RCTs used instruments that were author-developed and reported no testing of validity or reliability [25, 34].

Table 1 Country, demographic data, and intervention details for RCTs

References	Country	Sex (*n*)	Age (mean ± SD)	Sport/s	Calibre	Peer-reviewed	Education curriculum	Intervention (type, duration, intensity)
Abood et al. [29]	US	F (30)	19.6 ± 1.0	Swim, soccer	College Division 1	Y	Total energy, carbohydrate, fat, protein, calcium, iron, zinc	F2F, 8 × 1 h over 8 weeks Control—supervised, study, no treatment
Abood et al. [30]	US	F (70)	19.0 ± 0.8	Swim, diving, x-country, athletics, softball, basketball, volleyball	College Division 1	Y	Female athlete requirements, macronutrient distributions, supplements, nutrition myths and beliefs, weight management	F2F, 2 × 1 h (of eight health education sessions) Control—supervised study, no treatment
Buffington et al. [38]	US	F (153)	NR	Soccer, swim, diving, x-country, athletes, basketball, volleyball, tennis, cheer	Division 1 US Air Force Academy athletes	Y	Energy balance, metabolic rate, making healthy choices, macronutrients, planning meals, recovery, strength training, healthy weight management, body image, body type, supplements	Online, 10 × 10 mins over 10 weeks Control—NR
Chapman et al. [39]	US	F (72)	NR	Softball	High school	Y	Ergogenic aids, dehydration, pre-competition meal, energy sources, supplements, weight control	F2F, 12 × 45 min lectures over 6 weeks Control—NR
Doyle-Lucas et al. [32]	US	MF (321) – NS	15.4 ± 0.1	Dance	Pre-professional	Y	Female triad, macro- and micronutrients, healthy eating habits, hydration, eating for health and performance	F2F (DVD), 3 × 1 h sessions over 3 days Control—no treatment
Patton-Lopez et al. [36]	US	M (238) F (297)	14.9 ± 0.91	Soccer	High school	Y	Hydration, pre-exercise fuelling, during exercise fuelling, recovery nutrition, body composition and image, maintaining muscle and staying well, eating well while eating out	F2F, 7 h (4 h—sports nutrition education; 3 h—team building workshops) over 2 years Control—no treatment

Table 1 (continued)

References	Country	Sex (*n*)	Age (mean ± SD)	Sport/s	Calibre	Peer-reviewed	Education curriculum	Intervention (type, duration, intensity)
Rastmanesh et al. [35]	Iran	M (36) F (36)	30.0 ± 7.6	NR	Elite, paralympic	Y	Food pyramid, general nutrition, weight loss	F2F, 4 × 3 h session Control—no treatment
Bolles [37]	US	MF (56) C (54) [M (22), F (32)]	NR	Athletics	College	N (D)	Body composition, energy balance, macronutrients, micronutrients, relationships between foods, fad diets, eating disorders, supplements, meal planning, meal timing, portions, grazing	F2F, 12 × 90 mins over 6 weeks Control—NR
Brown [31]	US	MF (38) C (32) [M (6), F (26)]	19.5 ± 1.3	Soccer, x-country, athletics, softball, tennis, baseball, equestrian, wrestling	College	N (T)	Fuelling, energy requirements, protein, carbohydrates, fats, hydration, ergogenic aids, pre-competition nutrition, recovery, vitamins and minerals, supplements, weight loss and gain	F2F, 5 × 1 h over 5 weeks Control—no treatment
Hawk [33]	US	F (64)	15.3 ± 1.2	Soccer	Elite Division 1 adolescent	N (T)	Carbohydrates, protein, fat, micronutrients, hydration, supplements, meal timing, sport beverages	Handouts, 20–30 mins × 4 weeks Control—no treatment
Karpinski [26]	US	MF (92) C (76) [M (14), F (62)]	19.2 ± 1.2	NR	College	N (D)	Meal timing, energy requirements, carbohydrates, fats, protein, supplements, hydration, injury prevention, weight gain and loss	Online, nine modules × 9 weeks (self-paced learning) Control—9 × general nutrition articles
Oldham [34]	US	M (31) F (51)	14.9 ± 1.7	Swim	Competitive	N (T)	Food pyramid, hydration	F2F, 1 × 1 h session Control—NR

Table 1 (continued)

References	Country	Sex (*n*)	Age (mean ± SD)	Sport/s	Calibre	Peer-reviewed	Education curriculum	Intervention (type, duration, intensity)
Rennolds [25]	US	M (9) F (49)	19.6 (SD NR)	NR	College Division 1	N (T)	Female athlete triad	F2F, 1 × 15 min session Handouts only, 1 × F2F (1 × 15 min) + handout Control—no treatment

F female, *M* male, *MF* male and female, *C* completers, *NR* not reported, *NS* not specified, *SD* standard deviation, *Y* yes, *N* no, *T* thesis, *D* dissertation, *F2F* face to face education, *RCTs* randomised controlled trials

The most frequent educational topic covered in the interventions were macronutrients (carbohydrate, fat and protein), including roles of each, requirements for athletes, and food sources ($n = 8$) [26, 29, 31, 33, 35, 37–39]. Education regarding energy intake (e.g. energy requirements or energy balance) [26, 29–31, 37, 38], dietary supplements/ergogenic aids [30, 31, 33, 37–39] and general healthy eating/food groups [31, 32, 34, 35, 37, 38] were each included in 6 of the 13 RCTs. Most of the education interventions (10/13 studies) used face-to-face education methods [25, 29–32, 34–37, 39]. Average participant contact time was approximately 5 h (300 min) [range 15–1080 min]. Online ($n = 1$) and handout ($n = 2$) interventions were considered to be untimed [25, 26, 33]. Intervention session frequency varied from a single session up to yearly, and duration ranged from 10 to 180 min. Control groups usually received no treatment ($n = 8$) [25, 29–33, 35, 36] or details were not reported ($n = 4$) [34, 37–39]. One study gave the control group access to general nutrition handouts, but did not engage the group [26], and two studies placed controls in a supervised study hall for the same amount of time the intervention group was receiving education [29, 30].

The weighted average pre/post-test nutrition knowledge score for the intervention groups was 59.2 ± 4.7% (range 9.5–71.6%) and 78.4% ± 5.9% (range 30.8–91.0%), respectively, with a mean increase of 19.2 ± 1.6% (range 5.6–55.2%) and ES of 2.9 (large effect; range 0.55–17.05) from pre-intervention (Fig. 2). The weighted mean pre/post-test knowledge scores for the control groups were 54.2 ± 3.9% (pre-test, range 19.0–71.6%) and 55.9 ± 4.1% (post-test, range 13.2–71.6%), with an average increase of 2.2 ± 0.4% (range − 6.6 to 8.2%) and ES of − 0.36 (small negative effect). Only the RCT by Abood and Black [30] did not report a significant improvement in nutrition knowledge scores; however, the study was generally focused on 'health education', including self-esteem, performance pressure, stress management, and nutrition knowledge.

4.2 Single-Arm Pre/Post Design

Most studies (59.4%; 19/32 studies) utilised a single-arm pre/post-test study design; 11 were peer-reviewed and 8 were theses (Table 2). The study by Nascimento et al. [28] included two different populations (adults and teenagers), which were considered to be two different study conditions. Another study by Gould [27] tested two intervention conditions (cooking class plus handouts, and handouts alone). Overall, a total of 668 athletes were included in the 19 studies, of whom 55.4% were male ($n = 370$). The weighted mean for age was 17.6 ± 1.2 years ($n = 14$ studies), while the weighted attrition rate was 14.7 ± 1.4% (range 0.0–76.1%) (Table 4). Most studies (12/19 studies; 63%) were conducted in the US, with the remaining being conducted in

Table 2 Country and demographic data, as well as intervention details, for single-arm studies

References	Country	Sex (*n*)	Age (mean ± SD)	Sport/s	Calibre	Peer-reviewed	Education curriculum	Intervention (type, duration, intensity)
Martinelli et al. [40]	Italy	M (3) F (4)	21.6 ± 2.4	Rugby, water polo, basketball, American Football, shotput	University elite	Y	Fuelling, hydration, recovery, protein, supplements, dietary analysis	F2F, six sessions over 5 months
Nascimento et al. [28]—teenage	Brazil	M (11)	15.4 ± 0.4	Swim, athletics, volleyball, tennis, fighting, cycling, rowing, sailing, surfing	Elite, sub-elite, developmental	Y	Healthy eating, food groups, pre/post training meals, hydration, competition nutrition, individualised advice	F2F, one lecture and 4 × 60 min individual counselling
Nascimento et al. [28]—adult		M (15) F (6)	23.7 ± 0.5					
Reading et al. [43]	Canada	M (175)	14.9 ± 1.8	Ice hockey	Competitive	Y	Energy balance, carbohydrate, fluid intake	F2F, 2 × 1 h over 3 weeks
Valliant et al. [53]	US	F (11)	19.5 ± 1.0	Volleyball	College Division 1	Y	Individualised advice	F2F, 4 × individual consultations over 4 months
Yannakoulia et al. [42]	Greece	F (32)	20.5 ± 1.6	Dance	Dance school	Y	Eating disorders, nutrition for dancers	F2F, 12 × 2 h sessions over 12 weeks
Collison et al. [48]	US	F (60)	19.4 ± 1.2	Volleyball, tennis	College	Y	Energy requirements, fat, synthetic versus natural foods, fluids, vitamins and minerals, weight control, bone health	F2F, 2 × workshops over 2 weeks
Daniel et al. [41]	Brazil	F (10)	17.2 ± 0.9	Volleyball	State and regional level	Y	Healthy eating, sports nutrition, anxiety, antioxidants, energy balance, body image, healthy food choices	F2F, six meetings over 8 months
Kunkel et al. [19]	US	F (32)	NR	NR	College	Y	Food pyramid, serving sizes, healthy food choices, in- and out-of-season diets, pre-competition meals, weight management, snacks, eating out	F2F, group peer education meeting, weekly
Philippou et al. [44]	Cyprus	M (23) F (11)	15.2 ± 1.5	Swimming	National	Y	Mediterranean diet, sports nutrition for adolescents, supplements, reading food labels	F2F, presentation and supermarket tour over one half-day
Rossi et al. [52]	US	18 (M)	NR	Baseball	College Division 1	Y	Food preparation and storage, micro- and macronutrients, food sources of macro- and micronutrients, portion sizes, nutrient timing, hydration, supplements	F2F, Presentation (90 min) and 3 × 45 min small-group sessions over 12 weeks
Simpson et al. [45]	New Zealand	17 (M)	17.0 ± 0.7	Hockey	Elite	Y	Hydration, body composition, supplements, training diet, recovery, game-day nutrition	Electronic, fact sheet (1 × week) and 1–3 min video, ongoing feedback on app over 6 weeks

Table 2 (continued)

References	Country	Sex (*n*)	Age (mean ± SD)	Sport/s	Calibre	Peer-reviewed	Education curriculum	Intervention (type, duration, intensity)
Gould [27]	US	M (4) F (20)	NR	Soccer, athletics, volleyball, tennis	College Division 1	N (T)	Cooking lessons, meal preparation, portion control, individualised advice, macronutrients, balanced diets, food pyramid, meal planning, snacking, eating out	F2F, two cooking lessons + two newsletters Handout, two newsletters
Bongers [46]	US	F (11)	19.4 ± 1.1	Athletics	College	N (T)	Total energy, carbohydrates, fat, protein, hydration	F2F, 4 × 1 h sessions over 4 weeks
Chieppa [47]	US	F (49)	NR	Soccer, basketball, volleyball	College	N (T)	Supplements (protein, vitamin, sodium, zinc, iron, caffeine), hydration, energy distribution, macronutrients, milk myths, carbohydrate loading, pre-competition meals	Handout, 3 × newsletters over 3 months
DuBose [49]	US	M (82)	16.0 ± 1.1	Soccer, swim, x-country, basketball	High school	N (T)	Hydration	Online, 15-slide presentation
Hayman [50]	US	F (6)	NR	Softball	College Division 3	N (T)	Energy requirements, weight management, body composition, macronutrients, micronutrients, hydration	F2F, over 1 day
Hodges [51]	US	F (46)	19.8 (SD NR)	Soccer, athletics, basketball, volleyball, tennis, golf	College	N (T)	Calcium benefits, sources, recommendations for intake, importance for females	F2F, 20 min presentation, once-off
Schwartz [55]	US	M (11)	16.6 ± 1.1	American Football	High school	N (T)	Hydration, carbohydrates, protein, fat, weight gain, balanced diets, eating out	F2F, 8 × 30 min group discussion over 8 weeks + handouts, eight newsletters
Wali [54]	US	M (11)	20.1 ± 1.0	American Football	College	N (T)	Macronutrients, micronutrients, meal timing, weight management, supplements, hydration	F2F, 5 × 1 h sessions over 3 weeks

F female, *M* male, *SD* standard deviation, *NR* not reported, *Y* yes, *N* no, *T* thesis, *F2F* face-to-face education

Table 3 Questionnaire description, completers, attrition, knowledge assessment results and effect size for RCTs

References	Questionnaire description	V/R tests	Completers (*n*)	Attrition (%)	Intervention pre-score (%)[e]	Intervention post-score (%)[e]	Intervention change (%)	*p* value	Control pre-score (%)[e]	Control post-score (%)[e]	Control change (%)	Hedges' *g*
Abood et al. [29]	AD	PT, R, CV, TRT	30 (15 con, 15 int)	0	70.2 ± 1.3	76.2 ± 1.6	6.0	< 0.05	66.7 ± 0.74	65.5 ± 1.5	− 1.2	6.71
Abood et al. [30]	AD	PT, R, CV, TRT	70 (35 con, 35 int)	0	68.5 ± 42.1	79.9[a]	11.4	NS	63.7 ± 37.5	57.1[a]	− 6.6	NA
Buffington et al. [38]	AD	PT	148 (48 con, 47 COPE, 53 int)	3.4	63.1 ± 12.6	88.3 ± 13.5	25.2	< 0.001	71.7 ± 10.3	69.8 ± 14.8	8.1	1.29
Chapman et al. [39]	Mod (Werblow)	PT	72 (37 con, 35 int)	0	67.9 ± 4.4	76.3 ± 8.0	8.5	< 0.01	67.9 ± 4.7	67.0 ± 4.7	− 0.9	1.41
Doyle-Lucas et al. [32]	AD	PT, TRT, Cron α	210 (64 con, 146 int)	34.6	66.5 ± 1.2	91.0 ± 1.0	24.5	< 0.05	62.7 ± 1.8	70.9 ± 1.5	8.2	17.06
Patton-Lopez et al. [36]	Mod (Walsh)	Cron α	217 (64 con, 153 int)	61%	43.0 ± 15.0	50.8 ± 13.3	7.8	< 0.001	41.7 ± 13.5	42.3 ± 11.5	0.8	0.55
Rastmanesh et al. [35]	Mod (Barr, Zawila, Werblow)	PT, TRT, Cron α	72 (30 con, 42 int)	0	42.3%[a]	81.3[a]	39.0	< 0.01	43.1[b]	NR	NR	NA
Bolles [37]	AD, Mod (Zawila)	AV	54 (27 con, 27 int)	3.7	67.2 ± 6.6	72.8 ± 8.9	5.6	0.003	64.0 ± 5.4	66.1 ± 6.9	2.1	0.83
Brown [31]	AD	CV	32 (14 con, 18 int)	15.8	54.5 ± 11.5	86.3 ± 10.7	31.8	0.008	53.4 ± 9.7	52.1 ± 16.5	− 1.2	2.47
Hawk [33]	No mod (Zinn)	PT, Cron α	19 (10 con, 9 int)	70.3	9.5 ± 14.7	30.8 ± 12.8	21.3	0.008	19.0 ± 9.7	13.2 ± 8.5	− 5.8	1.57
Karpinski [26]	Mod (Reilly and Maughan)	PT	76 (37 con, 39 int)	17.4	53.3 ± 9.4	64.5 ± 9.8	11.2	0.001	53.0 ± 10.4	58.2 ± 8.0	5.2	0.70
Oldham [34]	AD	NR	82 (41 con, 41 int)	0	NR[c]	88.2 ± 11.1	16.6	< 0.0005	NR[c]	71.6 ± 13.9	NR[c]	1.30
Rennolds [25]—PO	AD	NR	49 (19 con, 14 PO, 7 HO, 9 PH)	15.5[d] 6.7	62.2 ± 12.9	77.3 ± 7.0	15.1	< 0.001	41.7 ± 32.7	41.5 ± 31.6	− 0.2	1.43
Rennolds [25]—HO				53.3	56.6 ± 18.5	66.2 ± 17.7	9.5	< 0.001				0.83
Rennolds [25]—PH				0	23.9 ± 18.4	79.0 ± 10.6	55.2	< 0.001				1.35

AD author designed, *Mod* modified pre-existing instrument, *No mod* used original instrument without modification, *V/R* validity/reliability, *PT* pilot testing, *R* readability, *CV* content validity, *TRT* test–retest, *Cron α* Cronbach α, *AV* assumed validity with no validity/reliability measures conducted, *NR* not reported, *con* control group, *int* intervention group, *COPE* cognitive behavioural therapy intervention, *NA* not applicable, *NS* not significant, *PO* presentation only, *HO* handout only, *PH* presentation + handout, *RCTs* randomised controlled trials

[a]Standard deviation not reported

[b]No post-test conducted on the control group

[c]No pre-intervention test conducted

[d]Total attrition

[e]Data are expressed as mean ± standard deviation

Study name	Std diff in means	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
Abood et al.(2004) [29]	6.900	0.963	0.927	5.013	8.786	7.167	0.000
Bolles [37]	0.841	0.284	0.081	0.285	1.398	2.963	0.003
Brown [31]	2.529	0.476	0.227	1.595	3.463	5.309	0.000
Buffington et al. [38]	1.309	0.220	0.048	0.879	1.739	5.964	0.000
Chapman et al. [39]	1.427	0.264	0.070	0.910	1.945	5.405	0.000
Doyle-Lucas et al. [32]	17.119	0.849	0.720	15.456	18.782	20.172	0.000
Hawk [33]	1.639	0.531	0.282	0.598	2.679	3.087	0.002
Karpinski [26]	0.702	0.236	0.056	0.239	1.166	2.970	0.003
Oldham [34]	1.320	0.244	0.059	0.842	1.797	5.415	0.000
Patton-Lopez [36]	0.664	0.152	0.023	0.366	0.963	4.362	0.000
Rennolds [25] - HO	0.859	0.458	0.210	-0.039	1.756	1.875	0.061
Rennolds [25] - PH	1.392	0.445	0.198	0.519	2.265	3.125	0.002
Rennolds [25] - PO	1.461	0.395	0.156	0.686	2.236	3.694	0.000

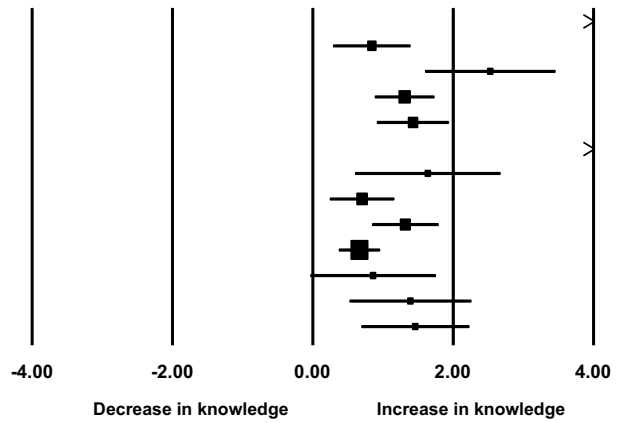

Fig. 2 Forest plot and effect sizes for randomised controlled trial studies. *HO* handout only, *PH* presentation and handout, *PO* presentation only

Italy ($n = 1$) [40], Brazil ($n = 2$) [28, 41], Greece ($n = 1$) [42], Canada ($n = 1$) [43], Cyprus ($n = 1$) [44] and New Zealand ($n = 1$) [45]. Eleven studies recruited populations of college- or university-level athletes [19, 27, 40, 46–48, 50–54], with four specifically recruiting Division 1 athletes [27, 52–54]. Two studies were conducted in high-school athletes [49, 55] and the remainder ($n = 6$) described their athletes as being elite national or international level, professional, or elite developmental [28, 41–45]. Volleyball athletes were most frequently studied ($n = 7$) [27, 28, 41, 47, 48, 51, 53]; however, ice hockey players made up the largest proportion of athletes ($n = 175$, 27%) in the single-arm pre/post design studies.

Nutrition knowledge instruments used by researchers were new ($n = 5$) [41, 42, 44, 50, 55], unmodified original instruments ($n = 4$) [19, 46, 53, 54] or modified from previously developed instruments ($n = 9$) [27, 28, 43, 45, 47–49, 51, 52]. Four studies conducted pilot testing for the study instrument [28, 44, 45, 48], while five reported no validation or reliability testing [40, 42, 50, 53, 55]. Three studies used part of another instrument plus other author-designed questions [28, 48, 49] and 10 studies used or modified a previously validated instrument without validation in their study population [19, 27, 43, 45–47, 51–54].

The curriculum of most single-arm studies included hydration ($n = 10$) [28, 43, 45–49, 52, 54, 55] and macronutrients (carbohydrates, protein, fat) ($n = 9$) [27, 43, 46–48, 50, 52, 54, 55]. Education regarding energy intake/requirements/balance ($n = 7$) [41, 43, 45–47, 50, 53] and micronutrients (iron, calcium; $n = 6$) [41, 46–48, 50, 52] was also common. Education was mostly delivered face-to-face (16/19 studies) [19, 27, 28, 41–44, 46, 48–55], but two studies used handouts or newsletters to facilitate nutrition education [27, 47]. One study used technology, distance-based education as part of the education intervention [45], and another utilised supermarket tours as an education medium [52]. Time spent educating athletes varied from 15 min (single presentation) up to 24 h (2-h sessions every week for 12 weeks); however, eight studies did not report the time participants were engaged in educational sessions [40]. Interventions ranged widely in duration (1 day–4 months) and frequency (weekly to monthly).

The nutrition knowledge-weighted mean pre/post-test scores across the single-arm studies were 64.4 ± 2.7% (range 18.4–73.6%) and 77.1 ± 3.9% (range 39.3–94.8%), respectively, which resulted in an average knowledge increase at post-intervention of 12.7 ± 0.8% (range 4.0–35.6%) and ES of 0.88 (large effect) (Fig. 3). Three studies found no significant improvements in nutrition knowledge post-intervention [41, 51, 55], and one study did not test if changes were significant pre/post intervention [50].

4.3 Combined Analysis (Randomised Controlled Trial and Single-Arm Studies)

When RCT and single-arm studies were combined ($n = 2180$), the majority of participants were female (66.1%; $n = 1440$). Of these, 1525 (70%) athletes completed the intervention, with a weighted average attrition of 20.5 ± 1.7%. Overall, the weighted mean age was 17.37 ± 1.3 years (range 10–32 years; $n = 23$ studies) [26, 28–37, 40–46, 48, 49, 53–55]. The majority of studies ($n = 24$; 75%) were completed in the US [19, 25–27, 29–34, 36–39, 46–55] and were using college- or university-age athletes ($n = 18$; 56.3%); seven studies specifically targeted Division 1 athletes [19, 25–27, 29–31, 37, 38, 40, 46–48, 50–54]. Six studies (18.8%) described their athletes as elite, or competitive internationally or nationally [28, 33, 35, 40, 44, 45], while only one study (3.1%) included athletes with

Table 4 Questionnaire description, *n* completers, attrition, knowledge assessment results and effect size for single-arm studies

References	Questionnaire description	V/R tests	Completers (*n*)	Attrition (%)	Intervention pre-score (%)[b]	Intervention post-score (%)[b]	Intervention change (%)	*p* value	Hedges' *g*
Martinelli et al. [40]	NR	NR	7	0	18.4 ± 12.0	39.3 ± 2.6	20.9	0.002	1.15
Nascimento et al. [28]—teenage	AD, Mod (Goncalves, Zawila)	DV, PT	11	NR	70.0 ± 9	89.0 ± 10.0	19.0	< 0.05	1.23
Nascimento et al. [28]—adult			21	NR	73.6 ± 15	86.4 ± 11	11.0	< 0.05	0.82
Reading et al. [43]	Mod (SNAC)	AV	33	81	44.2 ± 12.2	48.9 ± 10.4	4.7	0.001	0.50
Valliant et al. [53]	No mod (Reilly and Maughan)	NR	11	0	44.9 ± 10.7	57.3 ± 11.3	12.4	0.001	1.03
Yannakoulia et al. [42]	AD	NR	29	9.3	59.2[a]	94.8[a]	35.6	< 0.001	NA
Collison et al. [48]	AD, Mod (Werblow)	PT, R, CV, TRT	28	42.9	68.4 ± 9.4	90.0 ± 6.3	21.7	< 0.005	1.28
Daniel et al. [41]	AD	AV	10	0	57 ± 9.9	63.0 ± 11.8	6.0	NS	0.7
Kunkel et al. [19]	No mod (Werblow)	AV	32	0	66.7 ± 8.3	74.7 ± 9.6	7.7	< 0.05	0.99
Philippou et al. [44]	AD	PT	34	0	63.8 ± 21.0	69.7 ± 15.9	5.9	0.034	0.46
Rossi et al. [52]	Mod (Reilly and Maughan)	AV	15	16.7	56.7 ± 11.4	70.0 ± 9.4	13.3	< 0.034	0.79
Simpson et al. [45]	Mod (Parmeter, Werblow)	PT	17	NR	54.7 ± 14.3	61.1 ± 11.5	6.4	< 0.001	0.55
Gould [27]—cooking + newsletter	Mod (Werblow)	AV	11	0	74.9 ± 9.8	83.8 ± 6.9	8.9	0.01	0.92
Gould [27]—newsletter only	Mod (Werblow)	AV	13	0	70.5 ± 11.9	74.5 ± 10.1	4.0	0.004	0.40
Bongers [46]	No mod (Hoogenboom)	AV	11	0	74.7 ± 0.1	86.5 ± 0.1	11.8	< 0.05	1.36
Chieppa [47]	Mod (Corley)	AV	43	12	63.3 ± 10.6	69.8 ± 10.2	6.5	NS	0.73
DuBose [49]	AD, Mod (Rosenbloom)	AV	75	8.5	76.4 ± 8.7	91.8 ± 7.7	15.4	< 0.002	1.23
Hayman [50]	AD	NR	6	0	65.3[a]	88.7[a]	23.3	NR	NA
Hodges [51]	Mod (Hertzler and Frary)	AV	11	76.1	69.0[a]	76.0[a]	7.0	NS	NA
Schwartz [55]	AD	NR	11	0	68.0 ± 9.0	72.8 ± 6.8	4.8	NS	0.63
Wali [54]	No mod (Rosenbloom)	AV	12	7.7	69.4 ± 15.4	92.1 ± 7.6	22.7	0.018	1.13

AD author designed, *Mod* modified pre-existing instrument, *No mod* used original instrument without modification, *SNAC* sport nutrition for the athletes of Canada, *V/R* validity/reliability, *PT* pilot testing, *R* readability, *CV* content validity, *TRT* test–retest, *DV* discriminative validity, *AV* assumed validity with no validity/reliability measures conducted, *NR* not reported, *NA* not applicable, *NS* not significant

[a]Standard deviation not reported

[b]Data are presented as mean ± standard deviation

disabilities [35]. The most frequently studied sports, based on study number, were volleyball ($n = 9$ studies, 28.1%) [27, 28, 30, 38, 41, 47, 48, 51, 53] and soccer (football; $n = 9$ studies, 28.1%) [27, 29, 31, 33, 36, 38, 47, 49, 51], followed by track/athletics ($n = 8$ studies, 25%) [27, 28, 30, 31, 37, 38, 46, 51]. The most frequently recruited athletes, based on number, were soccer players ($n = 670$, 30.7%), ballerinas and dancers ($n = 242$, 11.1%) and ice hockey players ($n = 175$, 8.0%).

Study name	Std diff in means	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
Bongers [46]	1.414	0.477	0.227	0.480	2.349	2.966	0.003
Chieppa [47]	0.739	0.223	0.050	0.302	1.176	3.316	0.001
Collison et al. [48]	1.298	0.294	0.086	0.721	1.874	4.413	0.000
Daniel et al. [41]	0.733	0.462	0.213	-0.172	1.638	1.587	0.113
Dubose [49]	1.231	0.178	0.032	0.882	1.580	6.914	0.000
Gould [27] - CN	0.951	0.450	0.202	0.069	1.832	2.114	0.035
Gould [27] - NO	0.451	0.397	0.158	-0.328	1.229	1.134	0.257
Kunkel et al. [19]	0.999	0.265	0.070	0.480	1.519	3.769	0.000
Martinelli et al. [40]	1.226	0.583	0.339	0.085	2.368	2.105	0.035
Nascimento et al. [28] - T	0.836	0.314	0.099	0.220	1.453	2.660	0.008
Nascimento et al. [28] - A	1.278	0.468	0.219	0.361	2.195	2.731	0.006
Philippou et al. [44]	0.466	0.246	0.060	-0.016	0.948	1.896	0.058
Reading et al. [43]	0.508	0.250	0.063	0.018	0.999	2.033	0.042
Rossi et al. [52]	0.816	0.380	0.144	0.071	1.561	2.148	0.032
Schwartz [55]	0.654	0.438	0.192	-0.204	1.512	1.495	0.135
Simpson et al. [45]	0.560	0.350	0.122	-0.125	1.245	1.602	0.109
Valliant et al. [53]	1.071	0.456	0.208	0.177	1.964	2.348	0.019
Wali [54]	1.170	0.442	0.195	0.304	2.036	2.649	0.008

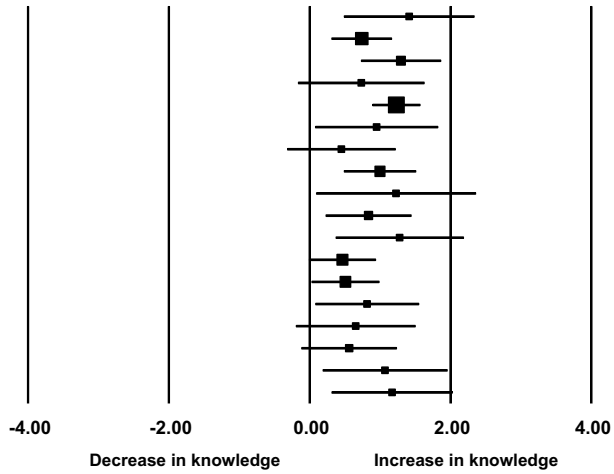

Fig. 3 Forest plot and effect sizes for single-arm studies. *CN* cooking lesson and newsletter, *NO* newsletter o nly, *T* teenager, *A* adult

The nutrition knowledge instruments used to assess athletes' nutrition knowledge were researcher-designed ($n = 12$ studies, 37.5%) [25, 29–32, 34, 38, 41, 42, 44, 50, 55], modified from a pre-existing instrument ($n = 10$ studies, 31.3%) [26, 27, 35, 36, 39, 43, 45, 47, 51, 52], existing instrument without modification ($n = 5$ studies, 15.6%) [19, 33, 46, 53, 54], or a combination of author-designed items and modified existing instruments ($n = 4$ studies, 12.5%) [28, 37, 48, 49]. Only one study reported no instrument validation details [40]. Pilot testing was the most frequently used form of validation/reliability testing ($n = 12$ studies, 37.5%) [26, 28–30, 32, 33, 35, 38, 39, 44, 45, 48]. Only 15.6% of studies ($n = 5$) utilised multiple (three or more) validation or reliability testing methods prior to using the instruments [29, 30, 32, 35, 48].

Thirty-six experimental intervention conditions were represented across all studies. The curricular items included in nutrition education interventions, by order of frequency, were macronutrients (carbohydrate, fat, protein; $n = 17$ studies, 53.1%), hydration ($n = 16$ studies, 50%), micronutrients (e.g. iron, calcium; $n = 12$ studies, 37.5%), weight management (gain/loss/maintenance; $n = 12$ studies, 37.5%), supplements ($n = 11$ studies, 34.3%), and general healthy eating (including food groups; $n = 11$ studies, 34.3%). Most (28/36, 77.8%) intervention protocols utilised some form of face-to-face education, with four including technology-based education (e.g. online or mobile applications) [26, 38, 45, 49]. Two studies included supermarket tours as part of their educational methods [36, 44].

The duration and frequency of nutrition education varied across studies, ranging from 10 min (single session) to 8 months (multiple sessions). Overall, intervention conditions (15/36 conditions, 41.7%) were short (< 4 weeks), and six of these were delivered as single sessions [25, 34, 44, 49–51] ranging between 15 min [49] to a half day [44]. Of these six, four reported a significant increase in knowledge from baseline [25, 34, 44, 49]. The most common length of each education session was 1 h or less (15/22 studies, where session length was indicated) and were delivered on a weekly basis (15/30 studies). Most intervention conditions ($n = 19$) had a total contact time of < 300 min (5 h). Five conditions, such as those educating only through handouts, were considered untimed.

The weighted average pre-test knowledge score was $61.5 \pm 1.8\%$, while the post-intervention score was $77.8 \pm 2.4\%$, an average increase of $16.1 \pm 0.7\%$. This represented a large ES of 1.72. One study failed to report whether the change in knowledge was significant [50]. Of the 35 remaining intervention conditions, only five (four face-to-face experimental conditions [29, 41, 51, 55] and one handout condition [47]) reported no increase in knowledge.

4.4 Assessment of Methodological Quality

All studies were assessed for methodological quality. For the RCT studies, the mean quality score was 17 ± 2.1 (range 14–19) out of a possible 28, while in single-arm studies the mean quality score was 11.7 ± 2.4 (range 6–16), with a maximum score of 22 (Electronic Supplementary Tables S1 and S2). All studies reported an hypothesis, and all but one study stated the main outcomes to be measured [45]. Overall, interventions were well-described, main study findings reported, and an estimate of random variability given (Electronic Supplementary Tables S1 and S2). Details related to funding or sponsorship were rarely reported and most studies did not adjust for confounding variables such as

previous nutrition education. Since most studies recruited based on a sport, team or sex, populations were generally not representative.

On average, the quality score for peer-reviewed research was 7.7% higher than theses and dissertations (Electronic Supplementary Tables S1 and S2). In peer-reviewed RCT studies, the mean quality score was 18.0 (range 15–19), compared with 15.8 (range 14–19) for theses and dissertations. For single-arm studies, the mean quality score for peer-reviewed research articles was 12.4 (range 6–16), compared with 10.9 (range 9–12) for theses/dissertations. Within RCTs (Electronic Supplementary Table S1), no study attempted to blind participants, main outcomes or conceal intervention assignment from athletes or facilitators.

5 Discussion

The purpose of this systematic review was to evaluate the effectiveness of different sport/general nutrition education strategies to improve nutrition knowledge in athletes. Overall, results show that nutrition education interventions (27 of 32 studies) significantly improved nutrition knowledge (mean improvement 16.1%; ES 1.72 [large effect]). Unfortunately, the range of education methods used makes it difficult to compare studies and determine which delivery method(s), including modality and dose, were most effective. In addition, the use of different sport/general nutrition knowledge assessment instruments, most with limited validation or reliability testing, makes it difficult to directly compare the changes reported and make inferences based on these changes.

Across all studies, athletes ($n = 2180$) were primarily female (66.1%), attending college/university (56.3%) and based in the US (75%). The bias towards female athletes should be considered since there is conflicting evidence on the effect of sex on nutrition knowledge level [14, 56]. Since the majority of athletes ($n = 18$, 58.1% of studies) represented in this review were athletes engaged in higher education, the improvement in knowledge observed may not be generalisable to younger or less well-educated athletes. Under-represented populations included elite athletes ($n = 6$; 18.8%) and those with physical disabilities ($n = 1$, 3.1%) [35]. Three studies compared nutrition knowledge between athletes of different calibres, but findings were equivocal [57–59].

Overall, interventions were primarily delivered in a face-to-face group setting, were short-term (typically < 4 weeks) and had a variety of session frequencies (mostly weekly) and durations (usually ≤ 1 h). The total contact time was generally < 300 min across the whole intervention. Long-term education programmes (> 5 months), which have been suggested to have the best efficacy at improving nutrition behaviours [1], have not been evaluated in athletes.

Handout-based interventions were not as effective as other education methods. Two studies comparing 'handouts alone' with another modality reported less knowledge improvement compared with handout plus cooking classes (4.9% difference), or presentation alone (5.6% difference) and a presentation plus handout condition (45.7% difference) [25, 27]. A previous study by Kicklighter et al. [60] reported that college students preferred an engaging facilitator over nutrition handouts for effective learning. Intervention delivery was mainly group-based using face-to-face presentations, which have been suggested to be more cost saving, time efficient, and as effective as individual counselling [61, 62]. Where there are large numbers of athletes, group education may indeed be the most viable teaching method, where educators such as registered dietitians (RD) are limited by funding or staffing capacity, as it facilitates active learning and peer interaction [63]. However, in a group setting, specific individual knowledge gaps, such as how to address food allergies/intolerances or selecting alternative foods to suit personal food preferences, may be challenging to address [64, 65]. In this review, studies that used individual consults did not appear to outperform group education studies; however, comparisons are difficult since no study compared both conditions concurrently. Further research into individualised education is also warranted as it can be tailored to target an athlete's weaknesses in knowledge.

Few intervention studies reviewed used technology-based (online or mobile) interventions, individualised consultations, supermarket tours, and/or cooking classes; thus, these educational approaches were under-represented compared with classroom-based interventions. Supermarket tours and cooking classes, which can teach practical, skill-based nutrition knowledge, has shown efficacy in their ability to educate participants on healthy food selection in non-athletic populations [66, 67]. Evaluation of these skilled-based education methods have reported high acceptability and feasibility, improved self-confidence, and social enjoyment [68–71]. Only four studies used technology-based education protocols [26, 38, 45, 49] that included online content or smartphone applications. When these approaches were used, attrition rates were lower (range 0–17.4%), while knowledge scores statistically increased (range 6.4–25.2%; ES 0.55–1.29). Athletes have significant time challenges due to busy training, competition, and travel schedules, and technology may therefore offer a viable means of supporting athlete education through a familiar medium (e.g. internet or smartphone). One study reported difficulty in scheduling one-on-one education sessions with athletes, which could be addressed by online delivery of nutrition education at their convenience [19]. Technology-enhanced learning is increasingly common, and research in non-athlete populations suggests

greater effectiveness of computer, internet, and multimedia-based education methods compared with paper-based resources [72–74]; however, this is relatively unexplored in athlete populations. Athletes, especially millennials and young adults, may have a preference for virtual professional support via e-mail, phone, mobile applications and social networks due to their high use of smartphones and the internet [75]. Jospe et al. reported that 32.4% of sport dietitians surveyed ($n = 180$) used mobile applications to monitor and track the dietary intakes of athletes due to the ubiquitous nature and convenience of smartphones [76]. Since health education programmes utilising technology show success in assisting the adoption of healthful behaviours in differing population groups, technology-based nutrition education programmes and tracking of dietary behaviours will continue to grow [77–80].

A major issue in comparing the effectiveness of different sport/nutrition education intervention programmes is the lack of validation of the knowledge assessment instruments used. This is consistent with previous reviews that identified inadequate validation as a major weakness within currently available studies [14, 56]. Without adequate validation of nutrition knowledge assessment tools, it is impossible to determine whether the tool effectively measures the construct and discriminates between athletes with stronger or weaker knowledge. Use of different instruments also makes it difficult to compare studies [81]. While the average sport/general nutrition knowledge improvement was 16.1%, the lack of validation of the knowledge instruments means it is difficult to infer how this improvement would translate to an athlete's practices. All but one study included in this review either designed a tool for the purpose of their study, with limited validation, or used a pre-existing questionnaire, often with some level of modification to make questions more appropriate for their study population. Several theses included performed no instrument validation [25, 34, 50, 55]. Knowledge questionnaires should be consistently updated, then revalidated to ensure they are up to date with nutrition recommendations and guidelines [82]. The development of more comprehensively validated instruments to assess sports nutrition knowledge, that are easily adaptable to different populations and food supplies, will allow a more robust comparison of different intervention modalities.

Overall, the studies informing this review were of average methodological quality (RCT, 17 ± 2.2 out of 28; single-arm, 11.7 ± 2.4 out of 22), with the majority of studies ($n = 19$, 59.4%) using a less robust single-arm pre- and post-test design compared with RCTs. If an RCT is not possible, future studies should consider the use of a stronger waitlist control design where athletes act as their own control before receiving the intervention. Control groups for RCT studies generally received no treatment, therefore the attention participants received was differential. In the future, the use of a sham control design (e.g. education unrelated to nutrition) is warranted to prevent attention bias [83]. Moreover, blinding of researchers conducting the analysis was not used in any study included in this review. Studies should consistently report all details of protocols, especially duration, frequency, curriculum and facilitator qualifications, as well as demographic details such as mean age, and calibre and measure of engagement such as attendance and attrition, to allow replication. Within the current literature, most intervention conditions [29 of 35] reported a significant and positive change in nutrition education knowledge, which suggests adequate power; however, power calculations were seldom reported.

5.1 Strengths, Limitations and Future Directions

This systematic review is the first to compare different sport/general nutrition education interventions for improving nutrition knowledge in athletes. The results provide informative direction for planning future education programmes, identify underutilised intervention types that warrant further investigation, and reveal a number of study design and methodological improvements that should be implemented in future research. A limitation of the included literature was the wide variance in intervention types and knowledge assessment tools that prevented identification of 'best practice' education method(s) and intervention intensity for athletes. One additional key consideration for future research is the issue of knowledge retention. Of 31 studies, only two conducted follow-up assessments to test for knowledge retention at 6 weeks [32] and 6 months [42], reporting minimal knowledge decreases. The effectiveness of education methods for helping athletes retain nutrition knowledge longer-term is essentially unknown. Additionally, the level of knowledge increase necessary to promote effective change in dietary behaviours is not well-established. Further research measuring long-term dietary change in comparison with change in knowledge requires additional research as this relationship is not well-established within the current athlete literature.

6 Conclusion

The heterogeneity of the current literature for sport/general nutrition education interventions makes it difficult to identify the most effective modality, duration, and frequency of education programming necessary to improve the nutrition knowledge of athletes. While the benefits of nutrition education in improving the nutrition knowledge of athletes is clear, the variability of knowledge assessment instruments combined with limited validity and reliability testing, as well as the wide range of intervention protocols, make it difficult

to draw conclusions for establishing best-practice guidelines in this area.

Future education programmes should first establish the validity and reliability of their nutrition knowledge assessment instrument, develop relevant and engaging programming to maximise participation, and test the suitability of the programme to the population being tested. Technology-based education modalities have not been well-explored in athletes, yet this approach may be attractive, cost effective, and engaging for athletes who are typically young millennials and heavy users of technology in other life areas, and increasingly using these technologies for monitoring training and athletic performance. Practical, skills-based nutrition education modalities including cooking classes and supermarket tours are currently under-researched within athletes despite success in other populations, and should be considered as part of future interventions.

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Compliance with Ethical Standards

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Conflict of interest Ryan Tam, Kathryn Beck, Melinda Manore, Janelle Gifford, Victoria Flood and Helen O'Connor declare that they have no conflicts of interest relevant to the content of this review.

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